COMPARATIVE RECIDIVISM RISK ASSESSMENT

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Meet Travis Power

22 years old, Male, African - American

Charged: Drug possession (non-violent felony)

Prior offenses: 1

Zero history of violence

The Score

COMPAS Called Him the Most Dangerous Person in the Room

COMPAS SCORE

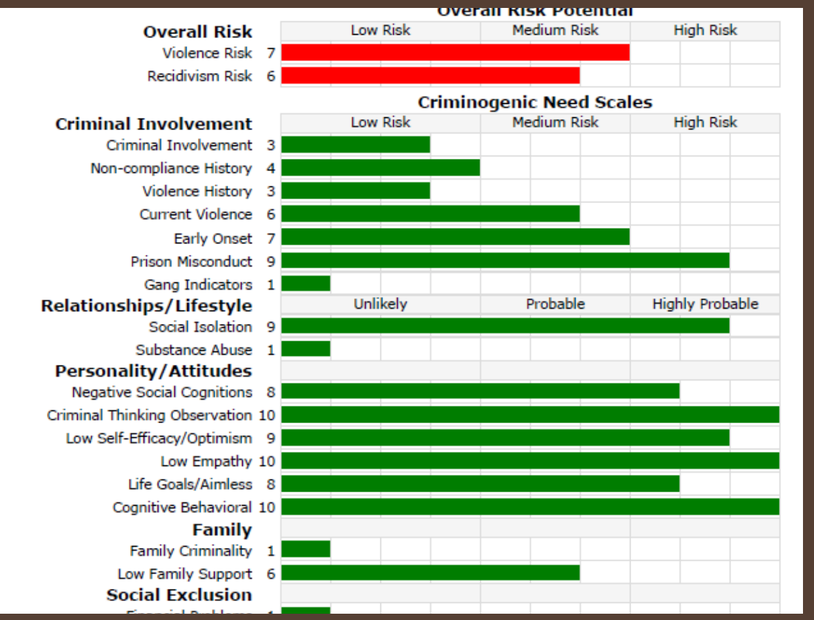
GENERAL RISK	VIOLENT RISK
10 HIGH RISK	HIGH RISK 10

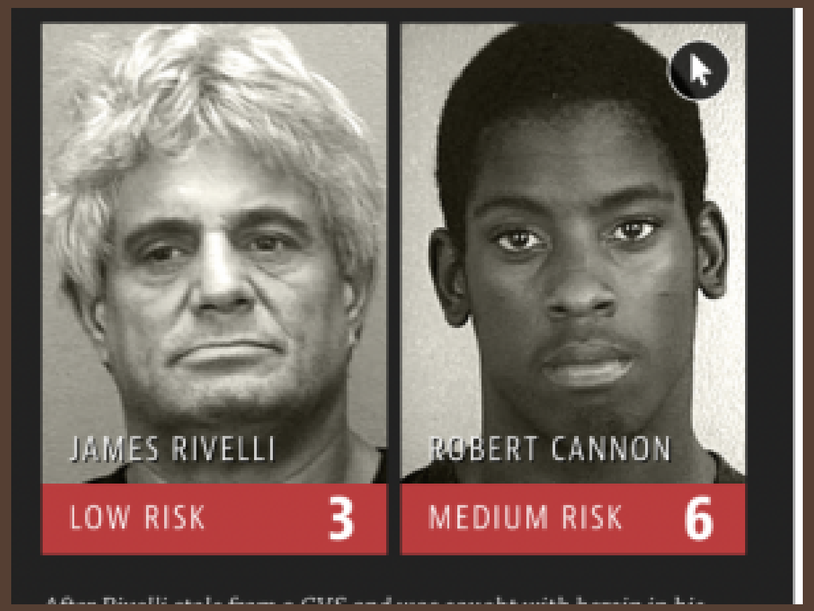
WHAT ACTUALLY HAPPENED

REOFFENSE	NON-VIOLENT
RESISTING ARREST	MISDEMEANOR

1 prior offense. No violent history. Maximum danger score — for both violence and crime.

COMPAS: A Score That Influences Sentencing





A questionnaire given to defendants

Produces a risk score from 1 (Low) – 10 (High)

Used by judges to decide bail, sentencing & parole

The Accuracy Problem

Wrong 1 in 3 Times

Over 2,000 people were misclassified

Some lost freedom they didn't deserve, others walked free when they shouldn't have.

The Bias Problem

The Mistakes Are Not Random

Same accuracy.
Completely different consequences depending on race.

🔴 **Innocent, but flagged dangerous**	🟢 **Guilty, but labeled safe**
Black defendants: 42%	**Black defendants:** 28%
White defendants: 22%	**White defendants: 50%**

JUDGE'S RISK TOLERANCE

Violent Defendants

Lock up 4 innocent people if we can **stop 1 violent criminal** from getting out.

Non-Violent Defendants

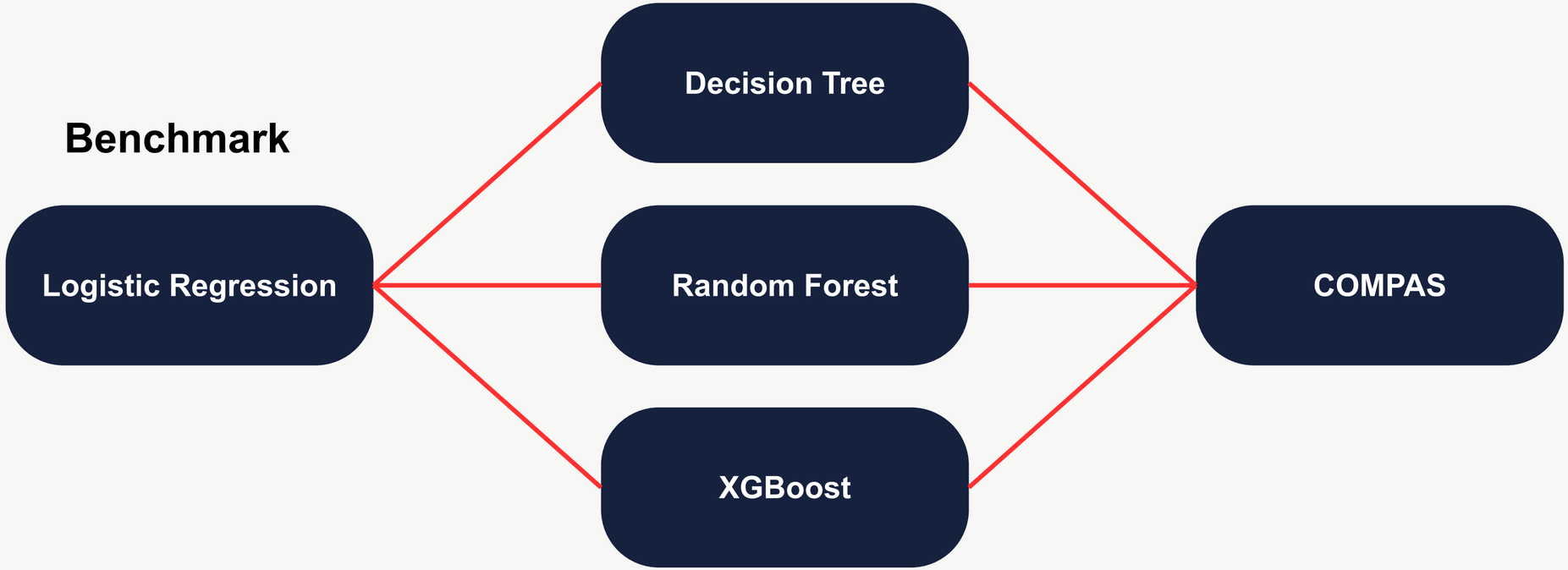
Only **lock up 1 innocent person** if we are **confident** enough to **catch 3 non-violent re-offenders.**

Case Type	Judge's Tolerance	Implied Threshold
Violent	4 FP per 1 caught	Low threshold — flag aggressively
Non-Violent	1 FP per 3 caught	High threshold — only flag if very confident

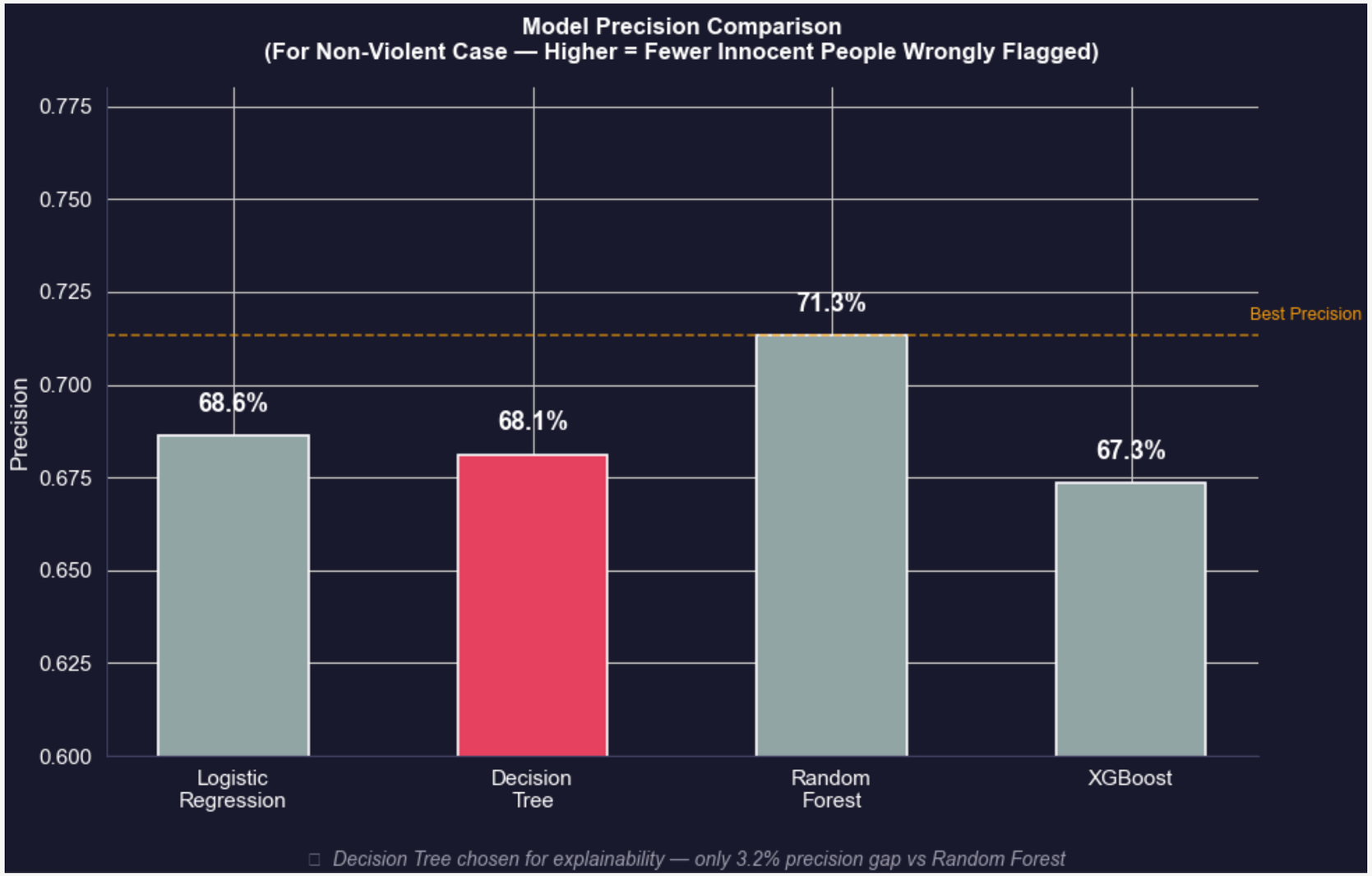
OBJECTIVE

Goal:

- Build a competing recidivism risk model that is **more transparent and fairer** than COMPAS
- Used it to make a recommendation for Travis Power's parole case.
- **Higher Precision, Low False Positive Rate**



MODEL COMPARISON

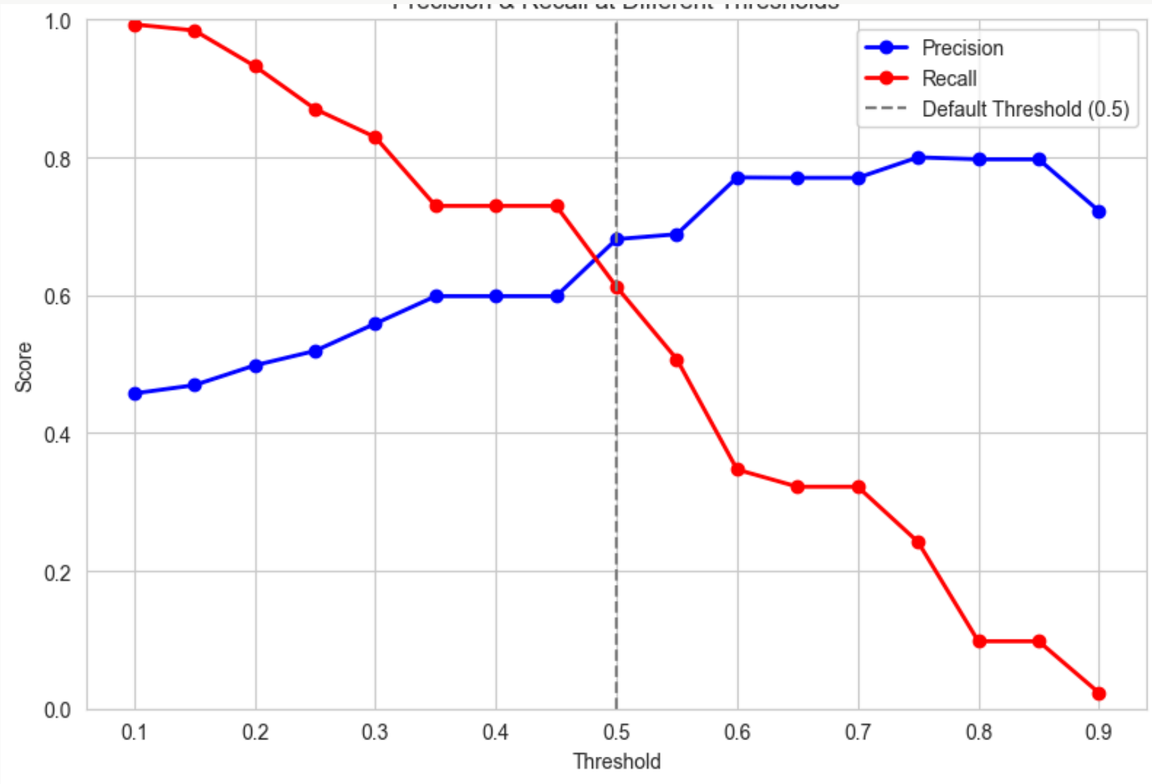


KEY INSIGHT

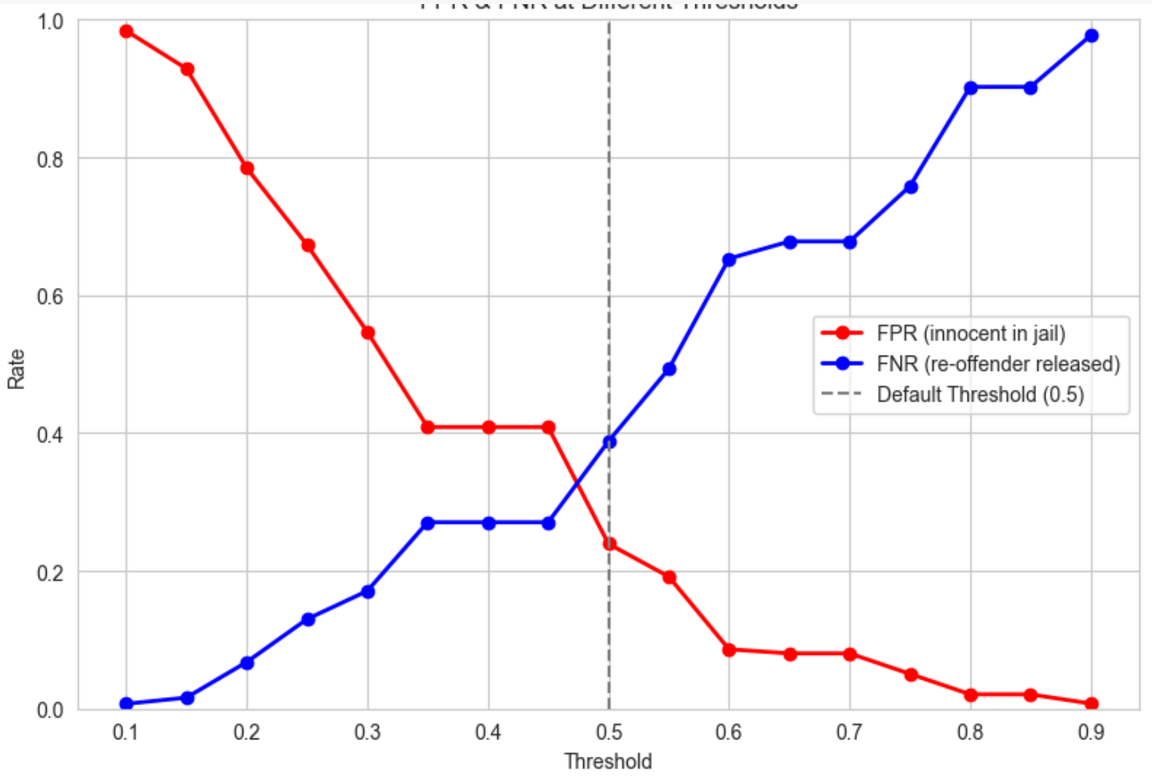
- **Random Forest wins** on precision (**71.3%**), but can't explain its decisions
- **Decision Tree chosen** — only 3.2% less precise, but fully transparent to the judge
- **All four models beat 50%** baseline — **every model adds real value**

THRESHOLD SELECTION (0.6)

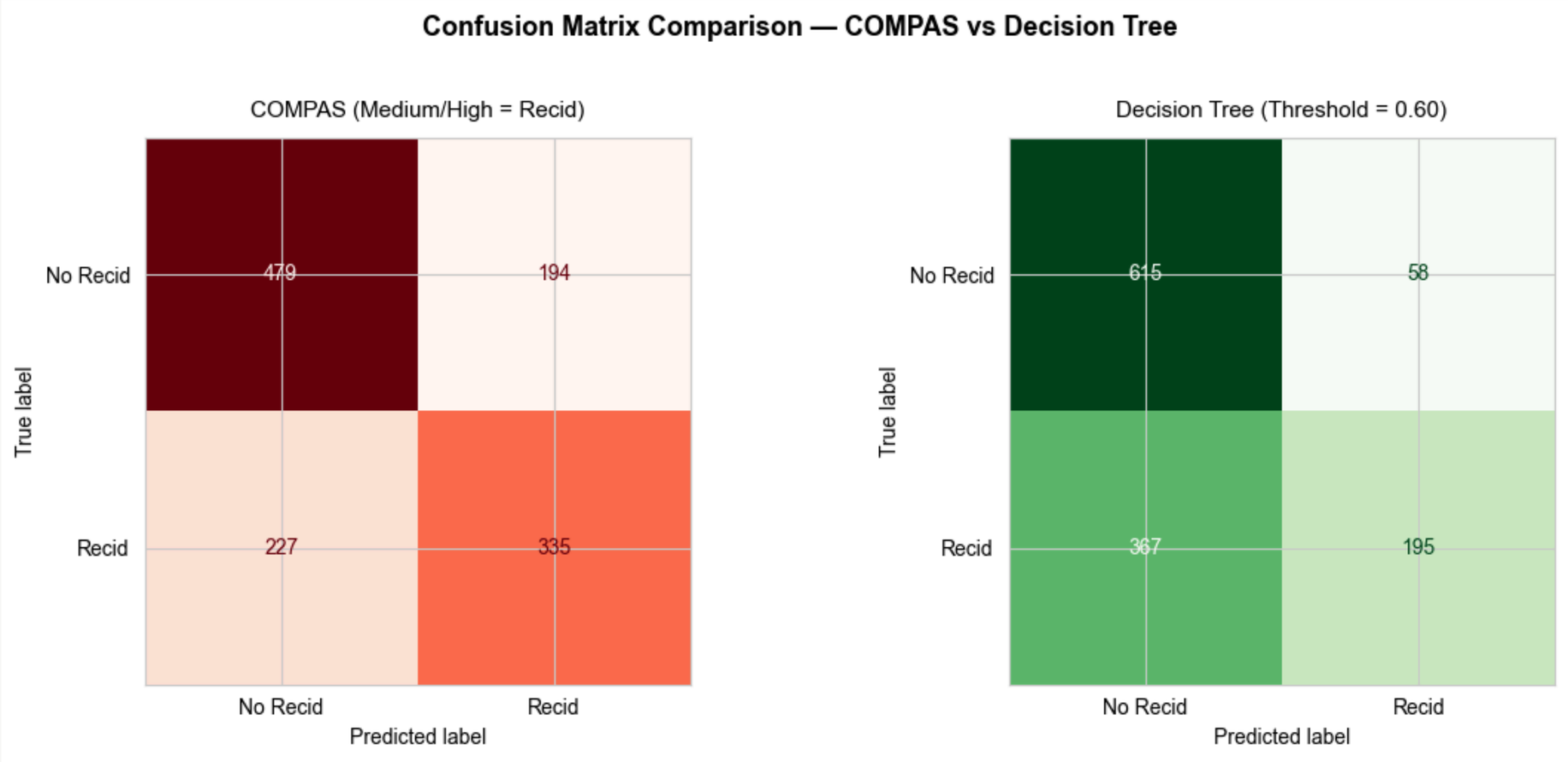
Precision & Recall at Different Threshold



FPR & FNR at Different Threshold



DECISION TREE VS. COMPASS

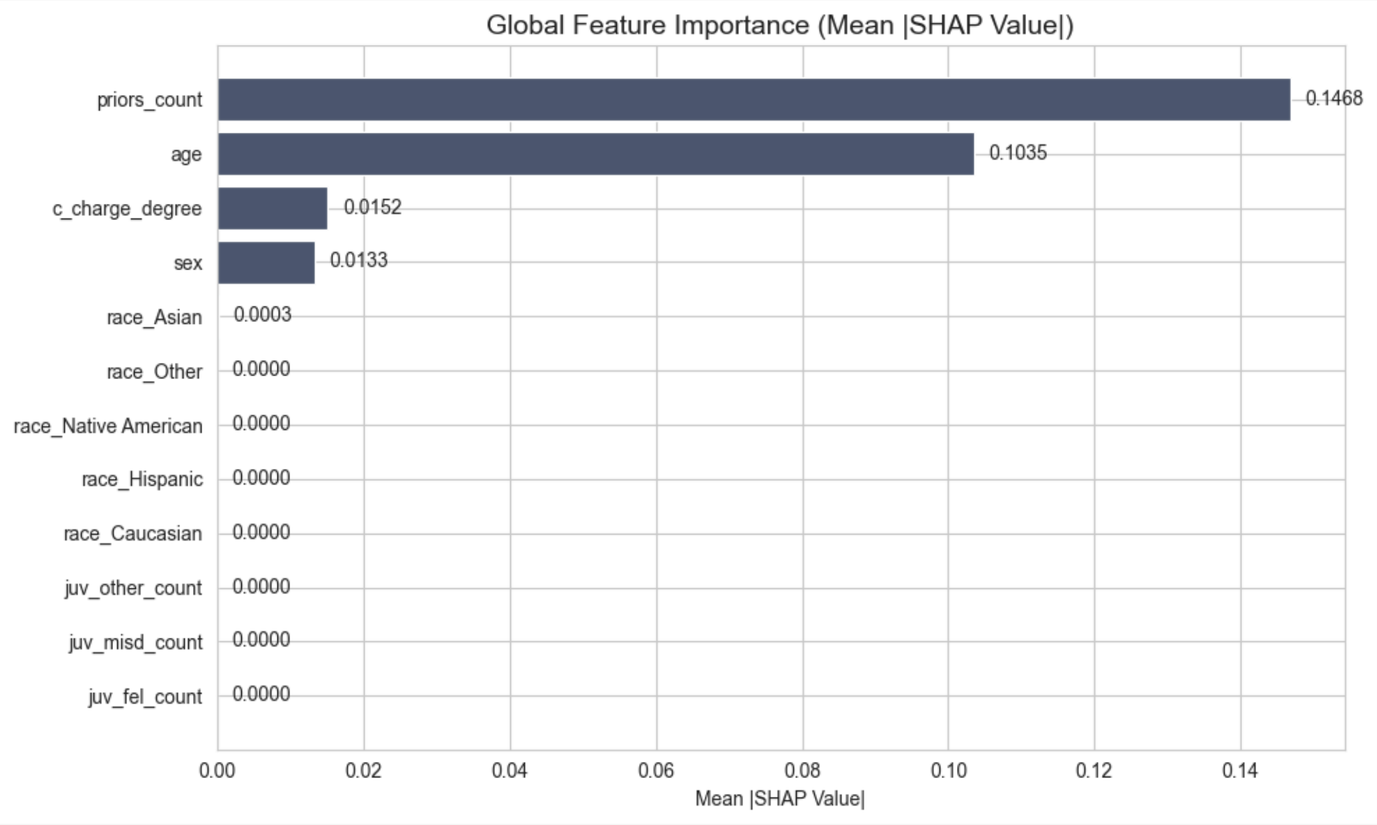


Precision = 65.9%
FPR = 29%
FNR = 40%

Precision = 77%
FPR = 9%
FNR = 65%

Model Interpretation

Feature Importance bar chart



Our model pays attention to:

- Priors
- Age

Race is irrelevant.

Model Interpretation

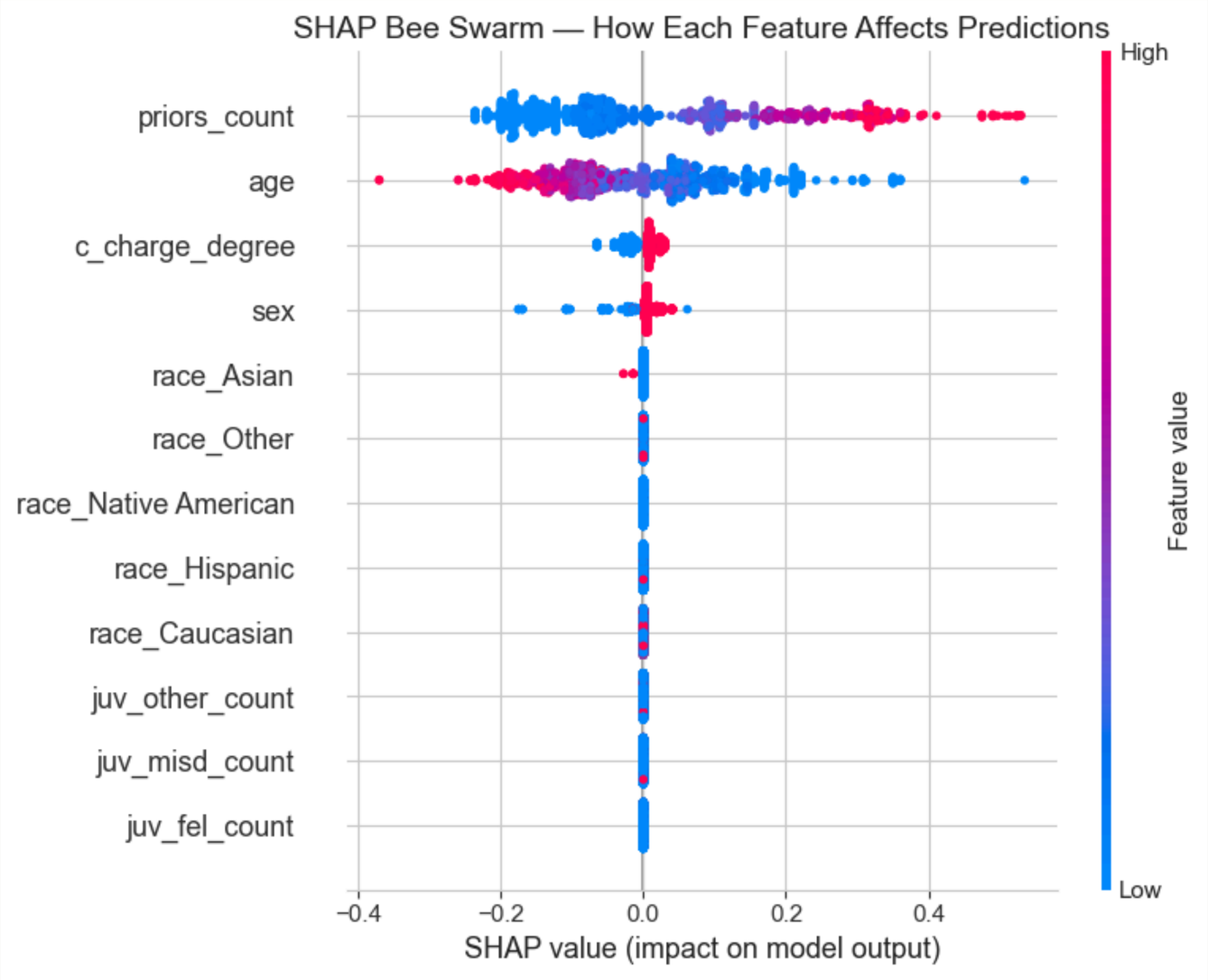
<u>Race wont affect prediction.</u>

01

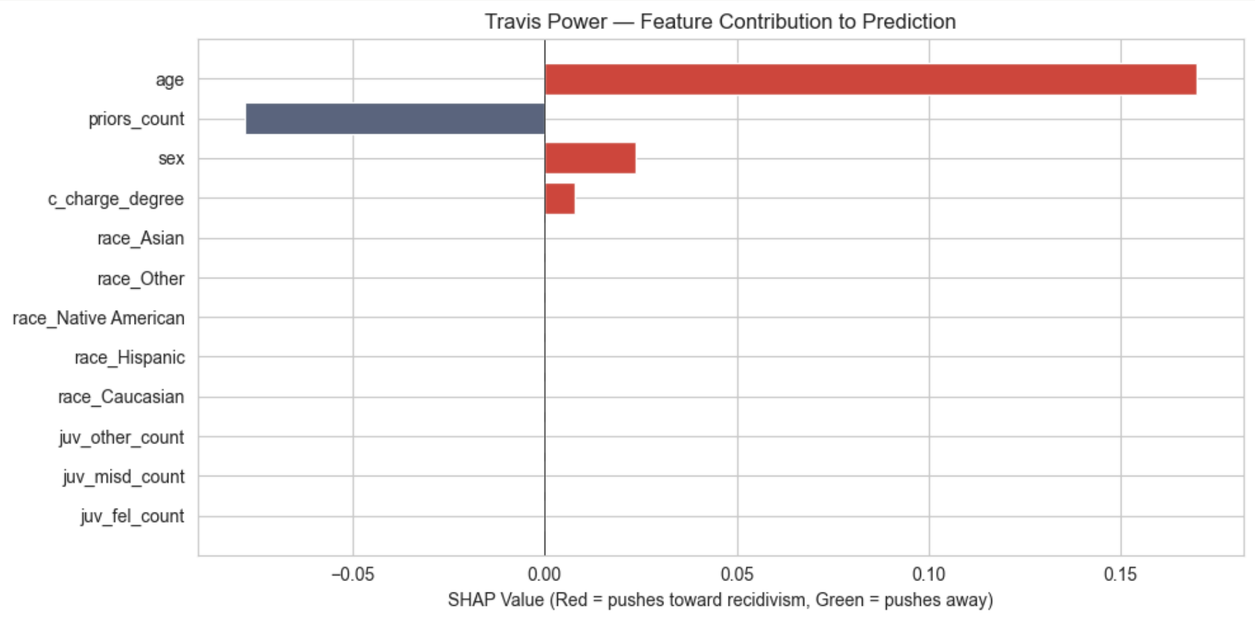
Only **priors** and **age** create a wide spread of dots.

02

Everything else, especially **race**, is bunched at zero.



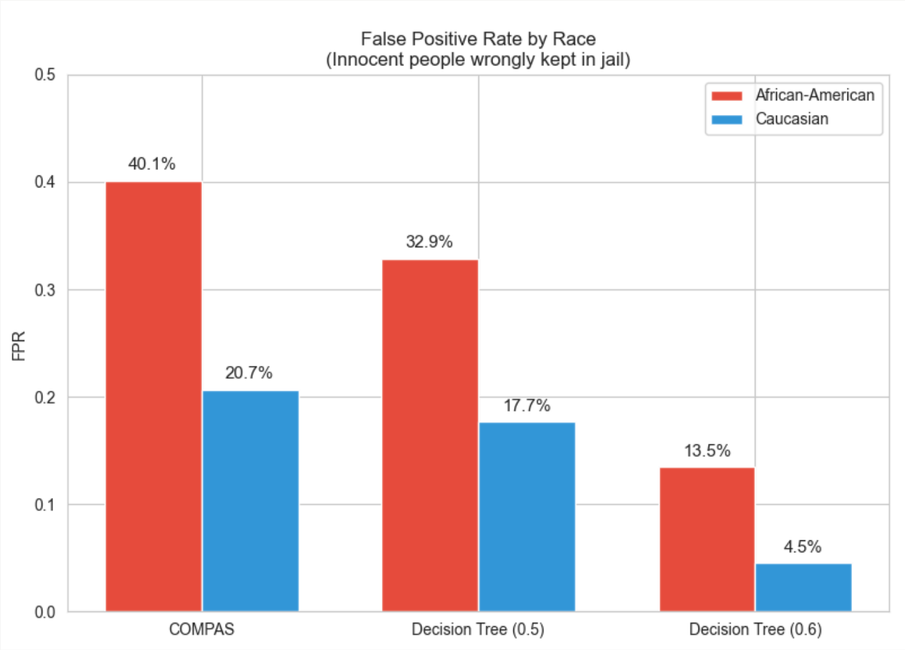
How Model Make Prediction

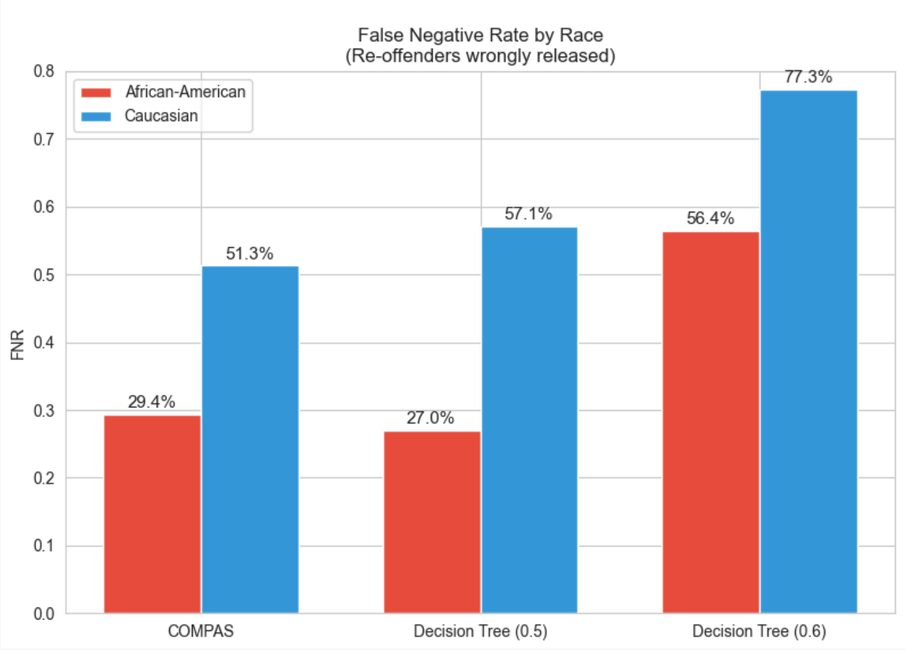


Now let's apply that to Travis specifically. Here's why he got his score

- Travis is **young**
- Travis has **few prior convictions**
- Being Black doesn't change his score

→ race has zero bar. The model doesn't see his race.





Fairness Analysis

Our model is significantly fairer to Black defendants than COMPAS

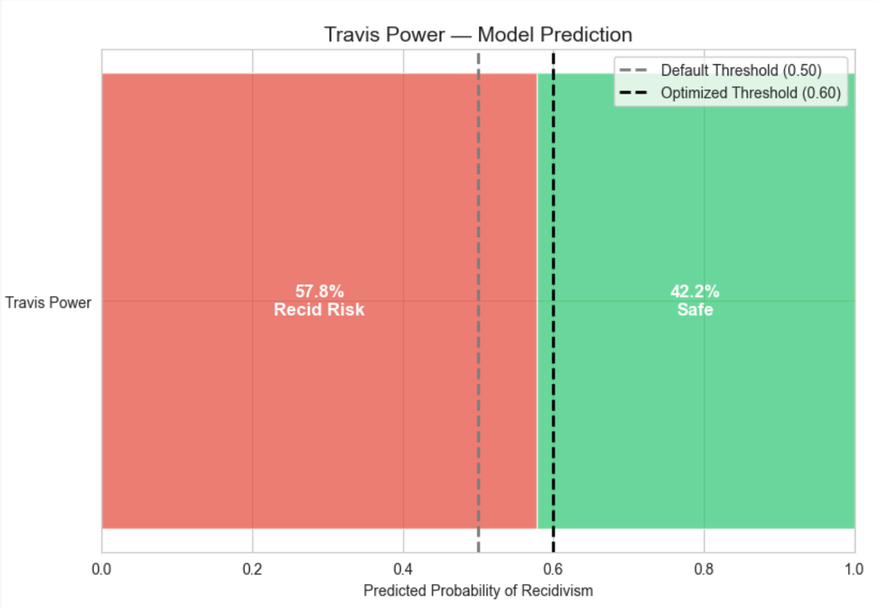
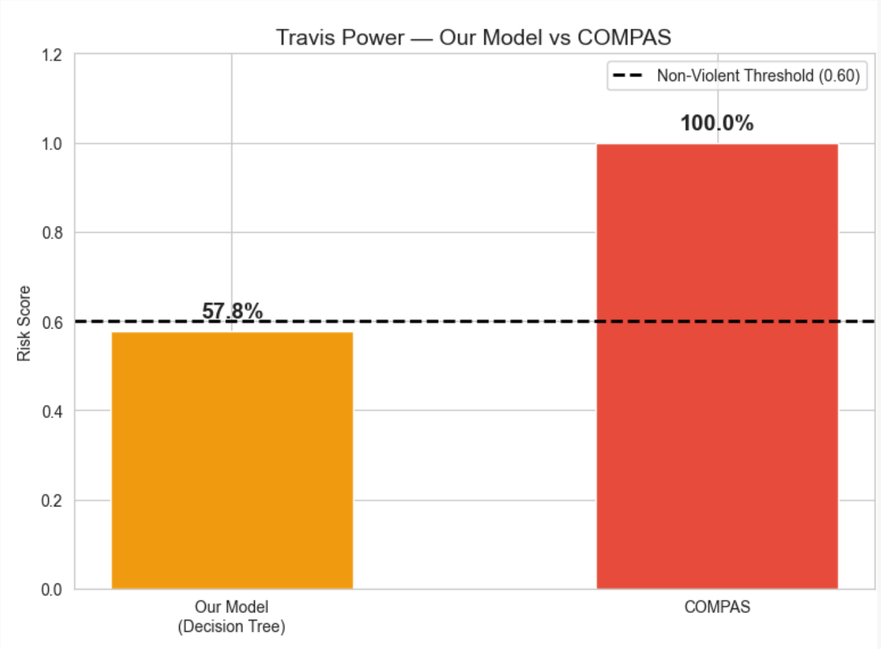
- COMPAS wrongly jails innocent Black people at **twice the rate**
- Our model cuts that injustice dramatically — from **40.1%** down to just **13.5%**
- **Yes**, we release more reoffenders — but that's the **right tradeoff** here

Final Recommendation

Travis scores below our safety threshold

- **We recommend granting parole**

- A **non-violent charge** and a **limited prior record** — the model is reflecting his reality.

- Travis scores **57.8%** — and our threshold is 0.6. He falls below the line we set for denial.
- His background supports that number.

→ **57.8%** is not a close call — it is a clear answer.